

DDSim has segmentation fault when running QAOA and Portfolio Optimiza

<https://github.com/cda-tum/mqt-ddsim/issues/248>

ROW 123

DDSim has segmentation fault when running QAOA and Portfolio Optimization with QAOA circuits from MQT Bench #248

New issue



Closed

#249

rentristandelacruz opened on Jun 14, 2023

mqtdsim version

1.17.2

OS

Linux Mint 21.1

Python version

3.10

C++ compiler

11.3.0

Additional environment information

No response

Description

I am simulating some circuits from the MQT Bench (<https://www.cda.cit.tum.de/mqtbench/>) using DDSim. Specifically, I am the using `qasm_simulator` backend and for the circuits I download a set of target-independent (compiler: qiskit) qasm files from the MQT Bench web page. I am currently using the DDSim `v1.17.2`.

When I recently run a QAOA or Portfolio Optimization with QAOA circuits (e.g. `portfolioqaoa_indep_qiskit_3.qasm`, `qaoa_indep_qiskit_3.qasm`) using the `qasm_simulator` backend I encounter segmentation faults. I did not encounter the segmentation fault issue with the other circuits from the set (i.e. Deutsch-Jozsa, GHZ, Grover's, etc).

I was simulating the same set of circuits a couple of months ago and I did not encounter the segmentation fault when running the QAOA-related circuits. Upon checking, I was using DDSim `v1.13.1`. I prepared different Python virtual environments and installed different versions of `mqtdsim`. The segmentation fault when running QAOA-related circuits started occurring in version `1.17.0`. Version `1.16.0` can run the QAOA and Portfolio Optimization with QAOA circuits without the segmentation fault occurring.

Expected behavior

No response

How to Reproduce

1. Create a Python virtual environment and install `mqtdsim v.1.17.0`.

```
• mkdir ddsim-1.17.0
• python3.10 -m venv ddsim-1.17.0
• source ddsim-1.17.0/bin/activate
• pip install --force-reinstall -v "mqtdsim==1.17.0"
```

2. Download one qasm QAOA circuit and one Portfolio Optimization with QAOA circuit (target-independent, compiler:qiskit) from MQT Bench page (<https://www.cda.cit.tum.de/mqtbench/>). e.g. `portfolioqaoa_indep_qiskit_3.qasm`, `qaoa_indep_qiskit_3.qasm`.

3. Create a Python script, used the ddsim `qasm_simulator` backend and simulate both qaoa and portfolioqaoa circuit. e.g. script:

```
import sys
from qiskit import *
from mqtdsim import ddsim
qasmFilename = sys.argv[1]
quantumCircuit = QuantumCircuit.from_qasm_file(sys.argv[1])
backendSimulator = ddsim.DDSIMProvider().get_backend("qasm_simulator")
myJob = execute(quantumCircuit, backendSimulator)
```

Assignees

hillmich

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Update dependencies
cda-tum/mqt-ddsim

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```
from mqt import ddsim
qasmFilename = sys.argv[1]
quantumCircuit = QuantumCircuit.from_qasm_file(sys.argv[1])
backendSimulator = ddsim.DDSIMProvider().get_backend("qasm_simulator")
myJob = execute(quantumCircuit, backendSimulator)
print(myJob.result())
```



rentristandelacruz added **bug** on Jun 14, 2023

rentristandelacruz assigned **hillmich** on Jun 14, 2023



hillmich on Jun 14, 2023

Contributor ...

Thanks for bringing up this issue. I can reproduce the segmentation fault locally but unfortunately I did not find the cause yet.

Some observations:

1. The circuit runs fine when passing the file to the c++ command line version.
2. The circuit runs fine in python when removing the `izz` gates.
3. In principle the `izz` should work as they do in `test_standalone_simulator.py` (`test_native_two_qubit_gates`)



hillmich mentioned this on Jun 14, 2023

[Actually Populate the Params list from Qobjs cda-tum/mqt-core#333](#)

hillmich added 2 commits that reference this issue on Jun 14, 2023

Actually Populate the Params list from Qobjs ([#333](#)) ...

Verified 1a889d6

Update dependencies ...

b3d6e2a

hillmich mentioned this on Jun 14, 2023

[Update dependencies #249](#)

hillmich added a commit that references this issue on Jun 14, 2023

Update dependencies ...

4d40e5d

hillmich closed this as **completed** in [#249](#) on Jun 15, 2023

hillmich added a commit that references this issue on Jun 15, 2023

Update dependencies ([#249](#)) ...

Verified fbfe86a



hillmich on Jun 15, 2023

Contributor ...

The latest commit to the main branch should fix the problem. Could you test this as well? I will add a new test for the fix later and create a new release.



  **hillmich** added a commit that references this issue on Jun 15, 2023

Update dependencies (#249) 

 **Verified** fbfe86a



hillmich on Jun 15, 2023

Contributor ...

The latest commit to the main branch should fix the problem. Could you test this as well? I will add a new test for the fix later and create a new release.



rentristandelacruz on Jun 15, 2023

Author ...

The update works. I no longer encounter segmentation faults when running QAOA-related circuits using the `qasm_simulator` backend. Thank you for the quick fix.

Some additional information: I'm running MQT Bench's circuits on the different DDSim backends including `hybrid_qasm_simulator` and `path_sim_qasm_simulator`. I'm encountering something QAOA-related again but this time when using `hybrid_qasm_simulator` and `path_sim_qasm_simulator` backends.

When running a QAOA circuit using the `hybrid_qasm_simulator` (in either `dd` or `amplitude` mode), the following log appears: `ValueError: Only StandardOperations are supported for now.`

When running a QAOA circuit using the `path_sim_qasm_simulator` (in any of the modes), the following log appears: `ValueError: DD for non-unitary operation not available!`

Is this expected behavior or is it a possible issue? (maybe related to `rz` gates again?) I can open a separate issue with replication procedure if you think this is a bug. Thanks!



hillmich on Jun 15, 2023

Contributor ...

The `hybrid_qasm_simulator` and `path_sim_qasm_simulator` only support standard operations, i.e. unitary operations. For the QAOA example you provided you should be fine with removing the measurements at the end.

In fact, the simulator should automatically remove the measurements at the end of the circuit but I guess the barriers are blocking the removal. Maybe something @burgholzer can also comment on?



burgholzer on Jun 15, 2023

Member ...

I am fairly sure that the barrier statements are the problem. IIRC measurements at the end are already removed automatically in both of these simulators. Should be an easy fix by just skipping over barrier statements.

@rentristandelacruz would you mind creating a separate issue for that?



rentristandelacruz on Jun 15, 2023

Author ...

I am fairly sure that the barrier statements are the problem. IIRC measurements at the end are already removed automatically in both of these simulators. Should be an easy fix by just skipping over barrier statements.

@rentristandelacruz would you mind creating a separate issue for that?

I created a separate issue for this.



Actually Populate the Params list from Qobjs #333

<> Code

Merged hillmich merged 1 commit into `main` from `fix-qobj-params` on Jun 14, 2023

Conversation 2 Commits 1 Checks 0 Files changed 1 +3 -0



hillmich commented on Jun 14, 2023

Collaborator

This PR fixes a bug where the parameters of parameterized gates are not saved when reading in circuits from `QasmQobjExperiments`. Previously the `params` list was always empty.

Related:

- [Migrate to Qiskit BackendV2](#) `mqt-ddsim#190`
- [DDSim has segmentation fault when running QAOA and Portfolio Optimization with QAOA circuits from MQT Bench](#) `mqt-ddsim#248`



Actually populate the params list from qobjs

3cd4867

hillmich added the `bug` label on Jun 14, 2023

hillmich requested a review from burgholzer 2 years ago

hillmich self-assigned this on Jun 14, 2023

Jun 14, 2023 at 3:22 PM GMT+2



github-actions bot commented on Jun 14, 2023

Contributor

Cpp-Linter Report ✓

No problems need attention.

Have any feedback or feature suggestions? [Share it here.](#)



Reviewers

burgholzer ✓

Assignees

hillmich

Labels

bug

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

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